

RTU Imp topic for 4th Semester

DMS :-

unit 1 Question based on inclusion and exclusion property in 4 marks and (10 marks)

Equivalence Relation

types of function

mathematical induction rule 4 marks and (10 marks)

unit 2 Propositional logic - totology , contradiction , quantifiers, question based on prove two propositional logic

CNF and DNF

Finite state machine

unit 3 POSET , hasse diagram - lower and upper bound

Lattice and it's property

bounded , complemented lattice and binomial theorem formula

recurrence relation solution for homogenous or non homogenous (10 marks)

generating function in (4 or 10 marks)

unit 4 Binary operation Alegabric properties - commutative, closure, associative, identity and inverse

group ,semi group and monoid for prove

Abelian group : prove 10 marks

Normal group : prove

ring field 10 marks

Isomorphism and homomorphism of group

unit 5 definition of different graphs

Hamiltion and eulerian graph

Isomorphism and homomorphism of graph

chromatic number (2 marks but imp)

Dijkshtra algorithm for shortest path (10 marks)

MEFA Imp topics

Unit 1 Deductive and inductive analysis, Scope and nature of economics, managerial economics, National income , GDP,

Unit 2 Demand and supply function, demand and supply elasticity, demand forecasting,

Unit 3 Cost and production function, Different types of cost, long run and short run in cost

Unit 4 Comparison between all market (oligopoly, monopoly, monopolistic, perfect)

Unit 5 Solvency , liquidity and turn over ratio, Balance sheet, short definition of all financial analysis

TC:-

unit 1 (Imp for 2 marks) Definition and Importance, Process of TC, Aspects(Competence) of TC, Linguistic Intelligence and Howard Gardner's theory, LSRW Skills, forms of TC, **Styles and Features of TC** ,

unit 2 technical Text types and Its process to read and write, **Note making and its types, guidelines for good technical writing**, document design process and information collection ,Types of Technical documents

unit 3 Grammar rules question related to tense or basic grammar(4marks), **Minutes of meeting(MoM)**,

unit 4 Types of Articles And Process of Technical Purposal, **Report and resume writing**,